# Figure 15 full-scan outcome

This is a historical result for source `116767bd3`, before the radial-root optimization in `75a0ced4e`. See the baseline and archive record the commit containing this report is not the measured revision.

The scan finished within its budget, with results for all 35 kinematic points. It did not reach 0.5% statistical precision across the full grid. Seven rest-frame points passed the declared reference-normalized precision goal for both polarizations. Moving-frame and near-light-cone points generally remain limited by Monte Carlo variance. The earlier near-light-cone longitudinal offset did not persist.

There is one major comparison anomaly at $\left(\frac{E}{T}, \frac{k}{T}\right) = (1.5, 0.5)$, exactly $E = 3k$. Its saved author reference is suspect because strict domain branches omit equality. This point is flagged for reference validation and excluded from the ordinary agreement summary below. A corrected reference has not been computed, so this investigation does not establish the causal size of that omission or resolve the point's physics.

## Run completion and coverage

The immutable executable uses commit `116767bd3ed30491a2b15559d228171d20082abe`. The scan began on 18 October 2026 at 19:19:29 Europe/Zurich and all compute exited on 19 October at 07:17:30, after 11 h 58 min 1 s. The original hard deadline was 07:19:29. Graceful interruption started two minutes before that deadline; the watchdog disarmed only after all compute had exited. No scan jobs remain.

Main allocations, including the author-reference workers, peaked at 60 workers. The six reference workers initially left 54 native workers; their completion released those CPUs to native integration. Separately authorized vacuum/profiling probes are outside this campaign accounting. No source changes or saved-state fingerprint changes were made to the production run.

The grid covers $\frac{k}{T} = 0, 0.5, 1, 1.5$ and $\frac{E}{T} = 0.25, 0.5, \ldots, 2$, excluding the exact light cone, with additional $E = k \pm 0.05$ points. Every point has valid final checkpoints. Samples per workspace range from 166,000 to 1,584,000; the total is 15.716 million workspace sampling points and 25.912 million native process evaluations. Rest-frame isotropy derives $T = L$ from the same trace integral; those are not independent measurements. All 70 physical component rows have zero final unresolved/nonfinite samples. This numerical-stability check does not establish Monte Carlo convergence.

## Precision reached

The controller's acceptance rule is $\frac{\sigma_{\mathrm{MC}}}{|\rho_{\mathrm{reference}}|} \leq 0.005$ with at least 100,000 samples and zero final unresolved evaluations. Agreement of the central value with the reference is a separate check. A 0.5% statistical error does not guarantee a central value within 0.5% of the target.

| $\frac{E}{T}, k = 0$ | Samples | MC error / target | Result |
|---|---|---|---|
| 0.25 | 200,000 | 0.4998% | Passed, T=L |
| 0.50 | 244,000 | 0.4988% | Passed, T=L |
| 0.75 | 334,000 | 0.4989% | Passed, T=L |
| 1.00 | 462,000 | 0.4994% | Passed, T=L |
| 1.25 | 642,000 | 0.4998% | Passed, T=L |
| 1.50 | 882,000 | 0.4997% | Passed, T=L |
| 1.75 | 1,172,000 | 0.4996% | Passed, T=L |
| 2.00 | 1,584,000 | 0.4797% of estimate | Reference unavailable |

The seven passed rest-frame results agree with their targets within 1.18 quoted Monte Carlo errors. The eighth rest point has sufficient provisional precision relative to its own estimate, but no independent target was obtained within the reference timeout.

The unmodified controller reports 15 accepted component flags: fourteen rest-frame rows plus transverse at $(1.5, 0.5)$. That transverse flag uses the suspect reference as its denominator: its MC error is 0.413% of that reference but 0.781% of its own estimate. It is not counted here as a successful 0.5% validation. The job at that point remained budget-stopped because its longitudinal component did not pass.

Some rest-frame errors are marginally above 0.5% when divided by their current estimate rather than the predeclared reference. Both ratios are retained in the original summary. Exactly ten component rows have strict error/estimate at or below 0.5%; eight are four passed rest integrations and two are the targetless $\frac{E}{T} = 2$ rest result. This distinction reflects the denominator definition, not a change to results or stopping policy.

Away from rest, even ordinary timelike points mostly remain above target. For example, $(1, 0.5)$ ends at 498,000 samples with T/L errors of 0.727%/0.937% of their targets. At $(1.25, 1)$, 408,000 samples give 1.33%/2.21%. No moving-frame point passes for both polarizations.

## Near-light-cone longitudinal result

Values below are isolated NLO coefficients including vacuum and thermal contributions, with unit couplings and the electromagnetic charge factored out; errors are one Monte Carlo standard error, without a propagated author-code quadrature uncertainty.

| $(\frac{E}{T}, \frac{k}{T})$ | Samples | Native L | Reference L | Pull | Error / target |
|---|---|---|---|---|---|
| $(1.05, 1)$ | 250,000 | $0.00949367 \pm 0.00191959$ | 0.01211950 | −1.37 | 15.84% |
| $(0.95, 1)$ | 170,000 | $0.13832947 \pm 0.10321419$ | −0.01059693 | +1.44 | 974% |

The above-cone transverse result is $0.04115141 \pm 0.00879869$ versus 0.05320425, also −1.37 errors from target, with 16.54% target-normalized uncertainty. The original longitudinal pilot offset of +2.37 errors is absent in this scan. Across 125 above-cone saved checkpoints, neither component exceeds two errors from target; longitudinal stays between roughly −1.72 and −0.76 after 100,000 samples. Neither component below the cone exceeds 1.45 errors.

These are compatibility statements under broad errors, not proof of convergence. Large individual weights still matter: longitudinal max-weight impact is 14.9% above and 68.8% below the cone. Rare contributions increased estimated uncertainty during the ladder, so smooth $\frac{1}{\sqrt{N}}$ forecasts are not reliable for these points. A chi-square near one does not remove this limitation.

## Reference anomaly and remaining agreement

At $(1.5, 0.5)$, the raw saved values are:

| Component | Native NLO, 528,000 samples | Author reference | Pull |
|---|---|---|---|
| T | $0.02130332 \pm 0.00016631$ | 0.04030709 | −114.3 |
| L | $0.02044940 \pm 0.00018202$ | −0.00399609 | +134.3 |

The author values were freshly computed with exact inputs and the recorded binary hash; this is not a stale-cache comparison. In the pinned upstream `src/6_.cpp`, lines 598 and 685 use `k0 > 3*k` and `k0 < 3*k`, with neither branch handling equality. The limiting domain is not generally empty at this equality. Neighboring grid points at $\frac{E}{T} = 1.25$ and 1.75 have native and author agreement within 2.15 errors. This is strong evidence for a reference-boundary problem, but a flank/limit evaluation or corrected reference is still required before attributing the entire discrepancy to it.

Excluding those two flagged component rows, all 62 available reference comparisons lie within 2.79 quoted Monte Carlo errors. Many of those errors are large, especially in spacelike regions, so this does not establish sub-percent accuracy there.

Independent T/L targets are missing for $(0.25, 1)$, $(0.25, 1.5)$, and $(2, 0)$. Each reference job hit its 1,800-second cap. The spacelike cases retained a vector result but lacked density completion, so no T/L projection is accepted. Their raw partial outputs remain in the archive.

## Figure 15 comparison and saved artifacts

The scan integrates the isolated NLO coefficient only. Its Figure 15 overlay's native markers use the author's analytic LO plus nine times native NLO, matching $g = 3$. Smooth comparison lines now use the original published Figure 15 vector paths, calibrated to the printed axes; gray dashed lines are the published LO curves. The fourteen curves contain fifteen original path segments and 4,187 vertices. The blue transverse curve retains its original clipping gap. No smoothing, new reference integration, or changes to the 64 displayed native values and error bars were applied. Three points without complete references remain unplotted, as in the prior overlay.

Published paths have graphical precision and retain the original figure's clipping, including in the full-range overlay. They are not substituted for the independently computed targets used for residuals or acceptance.

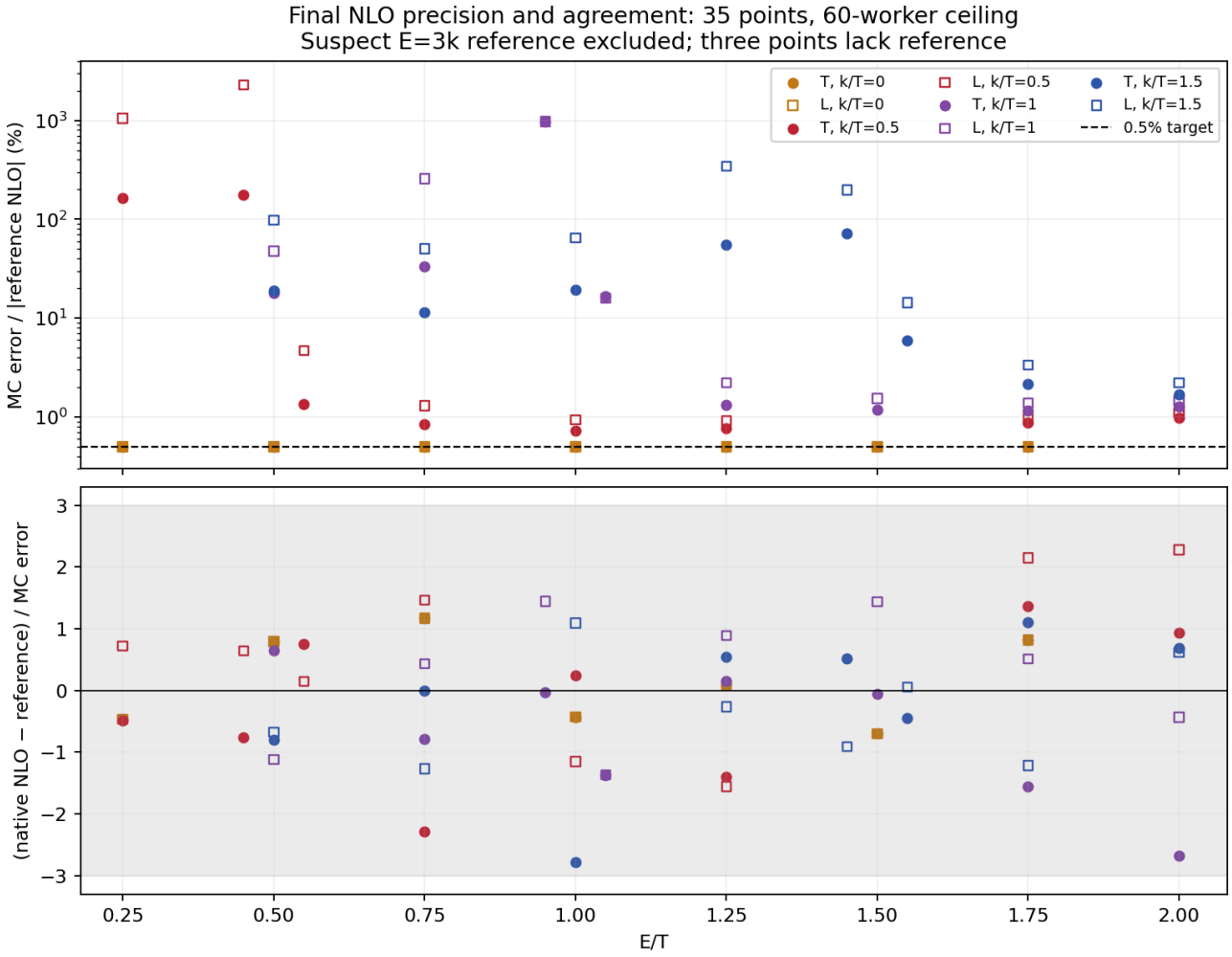



Figure 1: Final MC precision and reference agreement. Circles are T, open squares L. The suspected $E = 3k$ reference and missing targets are excluded explicitly; rest T/L overlap. The lower panel's ordinary residuals do not demonstrate small errors in the upper panel.

Full-range Figure 15 overlay and the fixed-range zoom show those published paths with the native markers. The zoom's transverse range begins at −0.4. The published curves do not show the suspect $E = 3k$ reference spike; the raw computed-reference discrepancy remains in the numerical diagnostics. The prior sparse-grid overlays are in the archive's checkout copy under `full-scan-20261019/rendering/recomputed-reference-grid/`. The zoom clips some native error bars; the full-range plot and CSV preserve those error bars. `../reference/published-curves.json` contains the calibrated paths; `figure15-render.json` records their provenance and all displayed markers. Physical NLO results, the LO-plus-NLO comparison, the interpreted audit, and the compact run provenance record the numbers and acceptance definitions. This manifest is explicitly curated; the original controller manifest is archived with its hash recorded here. Original summary flags are preserved even where this report qualifies their interpretation.

Consolidated native results retain all 35 final raw result objects. Author point results retain all 35 complete or partial source outputs, and the compact manifest retains runtime settings and seeds. The original per-slot stability records, checkpoint ladders, executed drivers, and complete generated states are in the verified archive described by the archive record, under the original checkout and `tmp/qcd-fig15-scan-20261018T171925Z/` paths. Archival and plotting perform no new integrations. No physics changes were made.

The next comparison that can resolve the strongest discrepancy is a one-sided author-reference check around $E = 3k$. Further native convergence work should focus on the high-variance signed-cut regions and their large weights, alongside the independently identified interval-root cost documented in the profiling report.